

Detection of Breast Cancer using Machine Learning Algorithms

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Abstract— Breast cancer is an abnormality in the growth of the breast and is one of the most common diseases found in women. The prompt diagnosis of any disease is very important to avoid further complications. Machine learning and deep learning algorithms play a vital role in diagnosis of disease like cancer. In order to detect breast cancer we developed a system in which the value of the breast cells from the patients report can be entered and the immediate response is obtained. This application gives a response about whether the cell is malignant or benign. This work is carried out on the Breast Cancer Wisconsin (Original) data set obtained from UCI repository. We have investigated five classification algorithms which include logistic regression, decision tree, random forest, KNN, and Naive Bayes for detection purposes. Among all these algorithms, the decision tree performs well obtaining the classification accuracy of 95.90%. Further, different feature extraction techniques can be employed to make the model more robust.

Keywords— Breast cancer, Decision tree, Machine learning.

I. INTRODUCTION

Breast cancer is the second most common cancer found in women in which the breast cells grow out of control. A lump in the breast, bloody discharge from the nipple and changes in the shape or texture of the nipple or breast are some of the symptoms in breast cancer. It can start in one or both the breasts. Breast cancer starts from different parts of the breast when the cells grow out of control. There were 2.3 million women diagnosed with breast cancer in 2020 and globally the deaths were 685000 [19]. In every country of the world breast cancer occurs in women at any age after puberty but with risks increasing in later life [19]. Breast cancer is widely studied throughout history and is the most commonly found disease in women. The doctors during the renaissance period saw revival of surgery. John Hunter, also known as the Scottish father of investigative surgery, identified lymph as a cause of breast cancer, it is a fluid carrying white blood cells throughout the body.

There are different types of symptoms for breast cancer but also in some cases there are no symptoms at all[18]. Some of the symptoms of breast cancer are given in table 1[17].

TABLE I. SYMPTOMS OF BREAST CANCER

1.	Lump in the breast or underarm
2.	Swelling of part of the breast
3.	Irritation or dimpling of breast skin
4.	Flaky skin or Redness around the breast or nipple area
5.	Pulling or pain in the nipple area
6.	Blood discharge from nipples other than breast milk.
7.	Changes in the size or the shape of the breast
8.	Pain in some areas of the breast

Due to the number of increasing cases in breast cancer, it has become tiresome for doctors to diagnose it. Hence, it is essential to have an automated system using which fast and accurate results of diagnosis of cancer can be obtained.

The aim of our system is to enter the patients data with ease and get the results of diagnosis within minimum time. Such systems will reduce the burden on doctors or clinical persons.

The next part of this paper is organized as follows: section 2 explains the related work presented in the literature which are studied by us. Section 3 elaborates on the materials and methods used in our study and section 4 demonstrates the proposed design and implementation details. Finally the results are discussed under section 5 followed by conclusion.

II. RELATED WORK

Machine learning and deep learning models have been used in detections of breast cancer in the past few years.

Many researchers have proposed different work for the detection of breast cancer using ML techniques. This section presents a few works related to the diagnosis of breast cancer.

A. Numeric dataset based work:

The work demonstrated [2], uses of data visualization and machine learning techniques like logistic regression, k-nearest neighbors, support vector machine, naïve Bayes, decision tree, random forest, and rotation forest. Even a comparative analysis using data visualization and machine learning applications for breast cancer detection was performed and diagnosis using Wisconsin Breast Cancer datasets (WBCD) with accuracy 98.1% for logistic regression algorithm.

The study demonstrated [5] focuses on how to deal with imbalanced data that have missing values to enhance the classification accuracy of detecting breast cancer using resampling techniques. In this work, two datasets and three classifier algorithms J48, NB, and SMO were used. The accuracy after preprocessing and resampling of J48, NB, SMO was 99.24%, 99.12%, 99.56% respectively using the Wisconsin Breast Cancer dataset (WBCD) while the accuracy for breast cancer dataset for algorithms J48, NB, SMO were 98.20%, 76.61%, 95.32% respectively.

The study [6] shows the work of six ML models, K- Nearest Neighbour, Logistic Regression, Decision Tree, Random Forest, Deep Learning Using Adam Gradient Descent using their hyperparameters for Wisconsin Breast Cancer Dataset were demonstrated in this paper. The supervised classification of malignant & benign cells is done by different ML algorithms and also by deep learning. The accuracy obtained by the Adam gradient Descent algorithm was the highest with 98.24% due to the combined benefits of AdaGrad and RMSProp.

The work [7], uses a rectified linear unit (ReLU) function, which did not create a vanishing gradient problem and also allowed the model to study faster and perform better. The algorithms used were Minimal Redundancy Maximal Relevance and Chi-square algorithms, the accuracy of Minimal Redundancy Maximal Relevance was 99.71%. This work can further be enlarged for breast cancer classification using medical images as these play a prominent role in the diagnosis of cancer using Wisconsin Breast Cancer datasets (WBCD).

B. Image dataset based work:

This paper [9] uses ML algorithms to suggest an approach for automated diagnosis of various types of breast carcinoma histopathological images. To choose the best features nondominated Genetic algorithm II (NSGA-II) algorithm and Ant Colony Optimization (ACO) were used. Also, to get the type of breast cancer two classifiers, probabilistic neural network (PNN) and multi-class support vector machine (MSVM) were put forward. Data was divided into a testing phase with 160 samples and 688 samples for the training phase. For the NSGA-II algorithm, the maximum accuracy obtained through the combined features set was 100% by training dataset and for PNN classifier 82.5% by testing dataset. Histopathological images database was also considered as a multi-classification task that used 848 BreakHis images with 400× magnification factors. NSGA-II algorithm produced an accuracy of 82.5% overall, 92.71% sensitivity and specificity of 97.57%.

TABLE II. SUMMARIZATION OF RELATED WORK

Author/ Year	Dataset used	Classification	Performance Measures
Numeric Based			
Muhammet Fatih AK et al. [2] (2020)	Wisconsin Breast Cancer datasets (WBCD)	Logistic regression algorithm	Acc. = 98.1%
Siham A. Mohammed et al. [5] (2020)	Breast Cancer dataset, Wisconsin Breast Cancer datasets (WBCD)	J48, SMO, NB	Accuracy Breast cancer dataset : J48= 98.20% SMO=95.32 % NB= 76.61% WBCD: J48= 99.24% SMO = 99.56% NB= 99.12%
Puja Gupta et al. [6] (2020)	Wisconsin Breast Cancer datasets	Adam Gradient Descent algorithm.	Acc.= 98.24%
Amin Ul Haq et al. [7] (2020)	Wisconsin Breast Cancer datasets	mRMR algorithm	Acc.= 99.71%
Image based			
Heba M. Afify et al.[9] (2020)	BreakHis database	Non- dominated sorting genetic algorithm II Algorithm	Acc=82.5% Sens= 92.71% Spec= 97.57%

III. MATERIALS AND METHODS

A. Classification algorithms:

This section explains the classification algorithm used in our work in brief. We have employed five classification algorithms for cancer detection purposes.

1) Logistic Regression:

Logistic Regression is a supervised classification algorithm that helps to predict categorical dependent variables based on the independent variable.

2) Decision Tree:

Decision tree is a supervised learning algorithm that is used for classification problems. In the Decision tree there are two types of nodes: the decision node and leaf node. Decision nodes make the decisions and may have sub nodes while leaf nodes are the result of these decision nodes and do not have sub nodes.

3) Random Forest:

Random forest is a collection of multiple decision trees but with different subset of given data and then averaged to improve the predictive analysis of the algorithm.

4) K-Nearest Neighbor:

K-Nearest Neighbor (KNN) algorithm is a supervised learning algorithm. This algorithm looks for similarities between the current case and available cases and then assigns the current case to a group more similar to it thereby helping to easily classify the data when visualized.

5) Naive Bayes:

This algorithm is a supervised learning algorithm which is based on Bayes' theorem. This algorithm is a probabilistic classifier in which it predicts based on the probability of the event.

B. Dataset used:

In this work, we have used Wisconsin Breast Cancer datasets (WBCD) obtained from UCI repository [20], [21]. This breast cancer database was obtained from the University of Wisconsin Hospitals, Madison from Dr. William H. Wolberg. This dataset has 699 instances. The dataset contains 11 attributes. The details of attributes are presented in Table III.

In the given dataset 458 instances out of total 699 belong to benign and remaining 241 belong to malignant class.

The distribution of benign and malignant class in the dataset is as follows:

Benign: 458 (65.5%)

Malignant: 241 (34.5%)

Total : 699

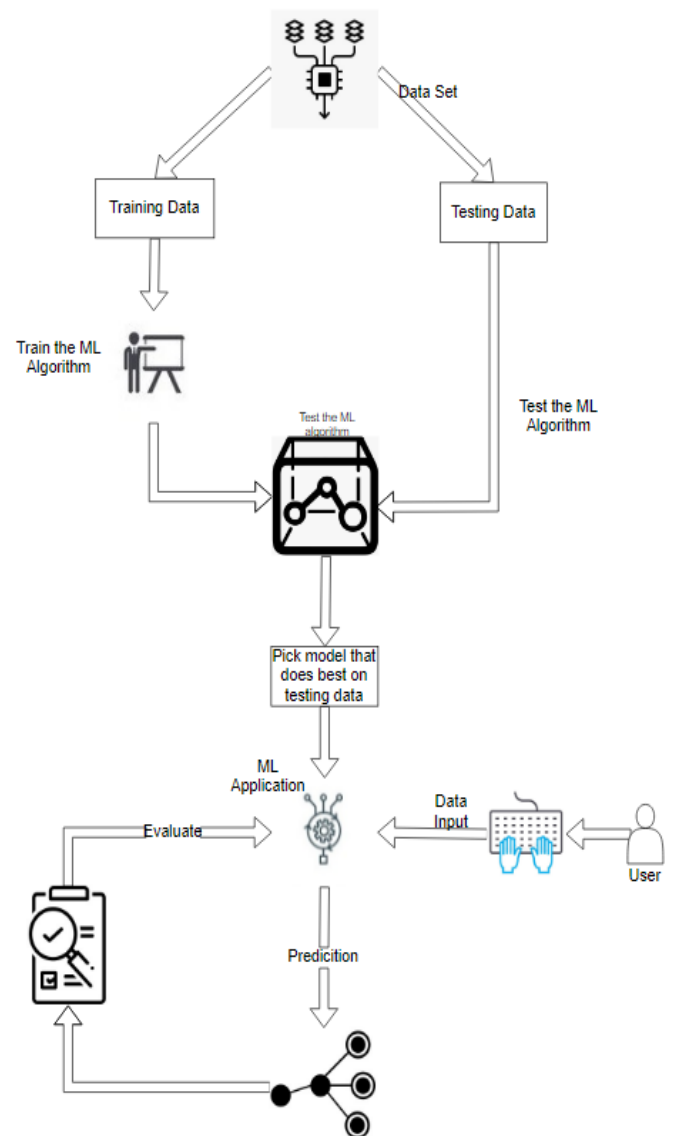
TABLE III. CELL ATTRIBUTES

#	Attribute	Domain
1.	Sample code number	id number
2.	Clump Thickness	1 - 10
3.	Uniformity of Cell Size	1 - 10
4.	Uniformity of Cell Shape	1 - 10
5.	Marginal Adhesion	1 - 10
6.	Single Epithelial Cell Size	1 - 10
7.	Bare Nuclei	1 - 10
8.	Bland Chromatin	1 - 10

9.	Normal Nucleoli	1 - 10
10.	Mitoses	1 - 10
11.	Class	2 for benign, 4 for malignant

IV. PROPOSED DESIGN AND IMPLEMENTATION

For any ML model to work the below-mentioned steps in the diagram are usually followed. First task will be the data collection after that data is preprocessed by removing any ambiguity and empty cells from the dataset. Further the model has to be trained on the input data and later it can be tested on the testing data. Different performance measures like accuracy, sensitivity, specificity and many more can be estimated to evaluate the performance of the model.



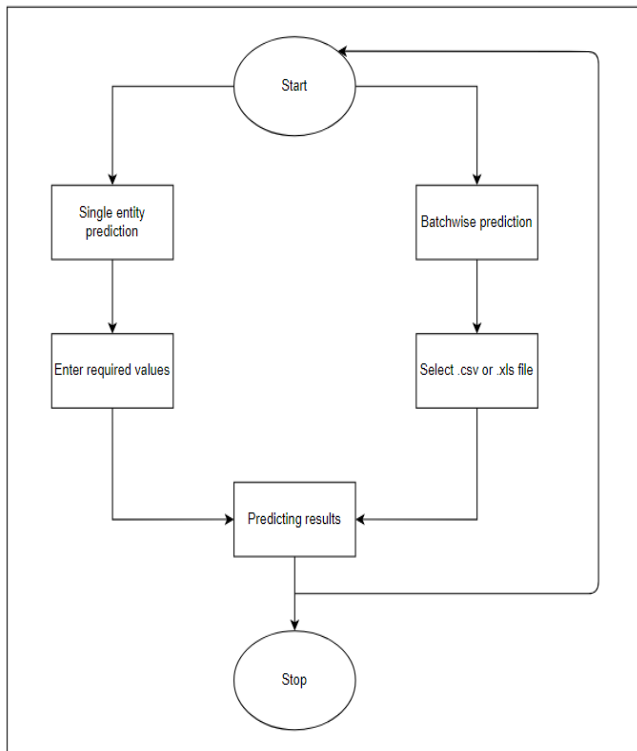
1. Fig. 1: System block diagram

Figure 1 shows the steps of implementation of the proposed work. To train the ML model the dataset is split into training and testing data. The dataset has been split in two parts, 75% for training and 25% for testing purposes. Logistic regression, decision tree, random forest, KNN, Naive Bayes

algorithms are used in the initial phase to check which algorithm gives better results on both training as well as testing data. The algorithm with the best results is selected to be used in the ML model.

Once the model is ready the system is ready to be deployed. Here when a user enters the cell value to the system, it predicts the result as whether the cancer is malignant or benign.

The novelty of this model is the new real time data entered will get stored in the dataset to increase the dataset volume which in turn would help increase the overall accuracy of the model. This will make the model more robust.



2. Fig. 2: Flow diagram for data entry

Figure 2 shows the flow for data entry in our system. The home page contains two options for data entry as a single entity and batchwise entity

For Single Entity Page:

Users need to manually enter the values of different attributes and select whether the result needs to be compared with a value or not.

For Batch Entity Page:

Users have to select a csv file by browsing through their device and the results of all entities will be printed in a tabular format saying if the cells are benign or malignant.

V. RESULTS AND DISCUSSION

Initially the model was trained with logistic regression, decision tree, random forest, KNN, Naive Bayes algorithms. The confusion matrix which helps in describing the performance of the classifier, has been determined for all five classifiers. Table 4 presents the confusion matrices for training and testing dataset. The performance measures considered for evaluation of these classifiers are accuracy, precision, sensitivity, specificity. The maximum testing classification accuracy obtained is using a decision tree

whereas the Confusion matrix is a N x N matrix used to represent the performance of the ML model. The columns of the confusion matrix represent the actual values while the rows represent the predicted values.

The performance measures used in this work are accuracy, precision, specificity and sensitivity. These parameters are defined as follows:

Accuracy is the measurement to determine which model is the best to identify relationships and patterns in a dataset between variables on the input or training data. It is the total percentage of rightly predicted values. The formula for calculating the accuracy is given as

$$\text{Accuracy} = (TP + TN) / (TP + FP + TN + FN)$$

The ratio between the true positives and all the positives is called precision, it also gives the measure of relevant data points. Precision is used to calculate the percentage of positive identifications that were actually positive. Precision is calculated by the formula

$$\text{Precision} = TP / (TP + FP)$$

Specificity is the proportion of actual negatives which got predicted as negatives or true negatives. Specificity stands for the proportion of total negatively identified values to the actual negative values.

The formula for specificity is given as

$$\text{Specificity} = TN / (TN + FP)$$

Sensitivity is the proportion of actual positives which are predicted as positives or true positives. Sensitivity value is used to calculate the total positives that are identified correctly.

Sensitivity is calculated by the formula

$$\text{Sensitivity} = TP / (TP + FN)$$

TABLE IV. CONFUSION MATRIX FOR TRAINING AND TESTING DATASET

ML algorithm	Confusion matrix for training dataset	Confusion matrix for testing dataset
Logistic Regression	M B M [332 5] B [5 170]	M B M [103 4] B [5 59]
Decision Tree	M B M [337 0] B [0 175]	M B M [104 3] B [4 60]
Random Forest	M B M [336 1] B [0 175]	M B M [104 3] B [5 59]
KNN	M B M [331 6] B [5 170]	M B M [102 5] B [5 59]
Naive Bayes	M B M [324 13] B [3 172]	M B M [100 7] B [3 61]

TABLE V. ACCURACY FOR TRAINING AND TESTING DATASET

ML algorithm	Accuracy for training dataset	Accuracy for testing dataset
Logistic Regression	98.04%	94.73%
Decision Tree	100%	95.90%
Random Forest	99.80%	95.32%
KNN	97.85%	94.15%
Naive Bayes	96.87%	94.15%

TABLE VI. PRECISION FOR TRAINING AND TESTING DATASET

ML algorithm	Precision for training dataset	Precision for testing dataset
Logistic Regression	98.51%	96.26%
Decision Tree	100%	97.19%
Random Forest	99.70%	97.19%
KNN	98.21%	95.32%
Naive Bayes	96.14%	93.45%

TABLE VII. SPECIFICITY FOR TRAINING AND TESTING DATASET

ML algorithm	Specificity for training dataset	Specificity for testing dataset
Logistic Regression	97.14%	93.65%
Decision Tree	100%	95.23%
Random Forest	99.43%	95.16%
KNN	96.59%	92.18%
Naive Bayes	92.97%	89.70%

TABLE VIII. SENSITIVITY FOR TRAINING AND TESTING DATASET

ML algorithm	Sensitivity for training dataset	Sensitivity for testing dataset
Logistic Regression	98.51%	95.37%
Decision Tree	100%	96.29%
Random Forest	100%	95.41%
KNN	98.51%	95.32%
Naive Bayes	99.08%	97.08%

According to the Table 5 Decision tree had the highest accuracy of 100% for training data so considering the decision tree we can see that decision tree also has the highest accuracy for testing data of 95.90%. The testing data had a total of 171 instances, of those 171 instances the decision tree identified 104 instances as TP i.e those values which were actually positive and were also identified as positive. 4 instances were identified as FN i.e those values which were actually positive but were identified as negative. The algorithm also identified 3 values as FP which were actually negative but were identified as positive. There were 60 values that were classified as TN. These were the values that were actually negative and the system also identified them as negative.

Taking into consideration these values that are derived from the confusion matrix we can calculate the accuracy, precision, sensitivity and specificity of the training and testing data.

VI. CONCLUSION

This work aims to detect breast cancer by classifying the breast cancer cells into benign or malignant cells. The dataset used for classification was the Wisconsin breast cancer dataset. In this implementation we have used a Logistic regression, Decision Tree, Random Forest, KNN, Naive Bayes classifiers. Out of these five algorithms the decision tree out performs with highest accuracy of 95.90%. Our system aims to help the doctors to examine the cells efficiently whether they are cancerous or non cancerous. In this system, doctors can enter the cell values specified in the patient's report with a given option for entering it and get a quick result about the patient's cancer condition. As a future work the model can be validated using the real time data of the patient. Even the cancer detection module based on image processing can be extended in future.

REFERENCES

- [1] Shler Farhad Khorshid, Adnan Mohsin Abdulazez, Breast Cancer Diagnosis Based On K-Nearest Neighbors: A Review, Palarch's Journal Of Archeology Of Egypt/Egyptology 18(4) (2021), 1927-1951.
- [2] Muhammet Fatih Ak, A Comparative Analysis of Breast Cancer Detection and Diagnosis Using Data Visualization and Machine Learning Applications, Healthcare 2020, 8, 111; doi:10.3390/healthcare8020111.
- [3] Ting Pang, Jeannie Hsiu Ding Wong, Wei Lin Ng, , Chee Seng Chan, Deep learning radiomics in breast cancer with different Modalities: Overview and future, Expert Systems with Applications, Volume158,15 November2020,113501.<https://doi.org/10.1016/j.eswa.2020.113501>
- [4] Jing Zheng, Denan lin, Shuang Wang,Mingjie He, Jipeng Fan, Deep Learning assisted Efficient Adaboost Algorithm for breast cancer detection and early Diagnosis, Special section on Deep Learning Algorithm for internet of medical things, 4 June 2020, DOI: 10.1109/ACCESS.2020.2993536
- [5] Siham A. Mohammed, Sadeq Darrab, Salah A. Noaman, and Gunter Saake, Analysis of Breast Cancer Detection Using Different Machine Learning Techniques, Springer Nature Singapore Pte Ltd, 2020 Y. Tan et al. (Eds.): DMBD 2020, CCIS1234,pp.108–117.https://doi.org/10.1007/978-981-15-7205-0_10
- [6] Puja Gupta, Shruti Garga, Breast Cancer Prediction using varying parameters of Machine Learning Models, Third International Conference on Computing and Network Communications CoCoNet' 19, Procedia Computer Science171 (2020) 593–601. 10.1016/j.procs.2020.04.064

- [7] Amin Ul Haq, JianPing Li, Muhammad Hammad Memon, Jalaluddin Khan and Salah Ud Din, A novel integrated diagnosis method for breast cancer detection, *Journal of Intelligent & Fuzzy Systems*, vol. 38, no. 2, pp. 2383-2398, 2020. DOI: 10.3233/JIFS-191461
- [8] Ghada Hamed, Mohammed Abd El-Rahman Marey, Safaa El-Sayed Amin, Mohamed Fahmy Tolba, Deep Learning in Breast Cancer Detection and Classification, *Proceedings of the International Conference on Artificial Intelligence and Computer Vision (AICV 2020)* pp 322-333. https://doi.org/10.1007/978-3-030-44289-7_30
- [9] Heba M. Afify, Kamel. K. Mohammed, Aboul Ella Hassanien, Multi-Images Recognition of Breast Cancer Histopathological via . Probabilistic Neural Network Approach, *Journal of System and Management Sciences*, Vol. 10 (2020) No. 2, pp. 53-68
- [10] Wenbin Yue, Zidong Wang, Hongwei Chen, Annette Payne, Xiaohui Liu, Machine Learning with Applications in Breast Cancer Diagnosis and Prognosis. *Designs* 2018, 2, 13; doi:10.3390/designs2020013
- [11] Li Shen, Laurie R. Margolies, Joseph H. Rothstein, Eugene Fluder, Russell McBride, Weiva Sieh, Deep Learning to Improve Breast Cancer Detection on Screening Mammography, *Scientific Reports* volume 9, Article number: 12495 (2019)
- [12] Susama Bagchi, Kim Gaik Tay, Audrey huong, Sanjoy Kumar Debnath, Image processing and machine learning techniques used in computer-aided detection system for mammogram screening: A review, *International Journal of Electrical and Computer Engineering (IJECE)*, Vol. 10, No. 3, June 2020, pp. 2336~2348. DOI: 10.11591/ijece.v10i3.pp2336-2348
- [13] Muhammet Fatih Aslan, Yunus Celik, Kadir Sabanci, Akif Durdu, Breast Cancer Diagnosis by Different Machine Learning Methods Using Blood Analysis Data, *International Journal of Intelligent Systems and Applications in Engineering Advanced Technology and Science*, 2018, 6(4), 289–293
- [14] Susama Bagchi and Audrey Huong, Signal Processing Techniques and Computer-Aided Detection Systems for Diagnosis of Breast Cancer: A Review Paper, January 2017 *Indian Journal of Science and Technology*, Vol 10(3), DOI: 10.17485/ijst/2017/v10i3/110640 January
- [15] Dana Bazazeh and Raed Shubair, Comparative Study of Machine Learning Algorithms for Breast Cancer Detection and Diagnosis, 5th International Conference on Electronic Devices, Systems and Applications, *IEEE Xplore*: 19 January 2017, DOI: [10.1109/ICEDSA.2016.7818560](https://doi.org/10.1109/ICEDSA.2016.7818560)
- [16] Mussarat Yasmin, Muhammad Sharif and Sajjad Mohsin, Survey Paper on Diagnosis of Breast Cancer Using Image Processing Techniques, *Research Journal of Recent Sciences*, Vol. 2(10), 88-98, October (2013)
- [17] https://www.cdc.gov/cancer/breast/basic_info/what-is-breast-cancer.html
- [18] <https://www.cancer.org/cancer/breast-cancer/about/what-is-breast-cancer.html>
- [19] <https://www.who.int/news-room/fact-sheets/detail/breast-cancer>
- [20] [https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+\(original\)](https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+(original))
- [21] William H. Wolberg and O.L. Mangasarian: Multisurface method of pattern separation for medical diagnosis applied to breast cytology, *Proceedings of the National Academy of Sciences, U.S.A.*, Volume 87, December 1990, pp 9193-9196.